

# Pedro Silva

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## EDUCATION

### Instituto Superior Técnico, University of Lisbon

MSc in Computer Science and Engineering

Lisbon, Portugal

Sep. 2025 – Aug. 2027 (expected)

### Faculty of Sciences, University of Lisbon (FCUL)

BSc in Informatics Engineering

Lisbon, Portugal

Sep. 2022 – Aug. 2025

## EXPERIENCE

### Software & AI Team Member

2025 – Present

AeroTec ATLAS, Systems & Aeronautics (IST UAV team)

Lisbon, Portugal

- Develop software for unmanned aerial vehicles in the only Portuguese university team building UAVs with autonomous flight capabilities
- Built Atlas Pipeline, a modular Python library for real-time video capture, processing and streaming (GStreamer) with ROS 2 integration and NVIDIA DeepStream AI inference for object detection
- Developed a ground-station camera system to command vehicles (minidrone and VTOL), receive live video feeds and monitor device status, with automated dev-environment setup scripts

### Programming Content Creator

2021 – Present

YouTube channel zFake

[youtube.com/c/zFake](https://youtube.com/c/zFake)

- Teach programming and computer science to a Portuguese-speaking audience, including a 2D game development series in vanilla JavaScript with open-source companion code

### Avionics Team Member, PalmaSat

2022

CanSat Portugal, 9th edition (ESERO)

Portugal

- Co-built the flight software of a can-sized microsatellite (telemetry, sensors, radio downlink); awarded the national **Technical Performance Award**

## PROJECTS

### Sophia | TypeScript, Node.js, discord.js, LLM tools, SQLite | [github.com/zF4ke/sophia](https://github.com/zF4ke/sophia)

- Built a production-grade conversational AI agent platform for Discord with 33 tools (moderation, role management, server search) driven by native LLM function calling in an agentic loop
- Engineered safety layers (approval gates, budget enforcement, context compaction, role-based access control), backed by deterministic and live test suites

### Timebox | TypeScript, Electron, multi-agent LLM systems, Vitest | [github.com/zF4ke/timebox](https://github.com/zF4ke/timebox)

- Built a multi-agent student planning app where LLM agents draft, critique and vote on weekly schedules under configurable quorum and iteration limits, with calendar (.ics) export
- Designed a benchmarking harness that runs a model/quorum/iteration matrix with pre-flight cost estimates, budget caps, per-call usage traces and deterministic mistake scoring

### Traveller | TypeScript monorepo, Next.js, MongoDB, discord.js

- Created a multiplayer RPG where cipher decryption is the core mechanic: platform-agnostic game engine, swappable persistence adapters, data-driven content pipeline and a Next.js admin dashboard, in a strict-TypeScript 5-package monorepo with an end-to-end suite that plays entire quests in under a second

### ModernBazaar | Java, Spring Boot, TypeScript, Next.js, PostgreSQL | [github.com/zF4ke/ModernBazaar](https://github.com/zF4ke/ModernBazaar)

- Built a market analyzer for a live game economy: Spring Boot ingestion backend, Next.js analytics frontend, real-time charts and historical trend detection over PostgreSQL

## TECHNICAL SKILLS

**Languages:** Python, TypeScript/JavaScript, Java, C/C++, SQL, HTML/CSS

**Frameworks:** React, Next.js, Node.js, Spring Boot, Tailwind CSS, ROS 2

**AI & Robotics:** PyTorch, OpenCV, GStreamer, NVIDIA DeepStream, LLM agents and function calling, neural networks

**Developer Tools:** Git, Linux, Docker, PostgreSQL, MongoDB, Firebase

**Courses:** Neural Networks: Zero to Hero (Andrej Karpathy), Node Basic (HackerRank), NLW IA (Rocketseat)

**Spoken Languages:** Portuguese (native), English (fluent)